

A REPLY TO TERENCE TAO'S OPEN LETTER "A SEVERE MISALIGNMENT OF AI IN MATHEMATICS"

Reflections relevant to linguistics, too

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The text of the open letter is available at:

<https://terrytao.wordpress.com/2026/09/11/a-severe-misalignment-of-ai-in-mathematics>

I see the concern about AI and pure mathematics somewhat differently.

First, I would distinguish not between solved and unsolved problems, but between problems on which substantial mathematical work already exists and new problems that have not yet been investigated. At present, AI seems particularly capable of consolidating and extending existing research. Whether it can formulate and solve mathematically significant new problems of its own is, in my view, a different question.

Second, a mathematical problem can admit more than one solution or proof, and different solutions may reveal different properties, structures, relations, etc.; crucially, different solutions may also differ in complexity (in the sense of Big O notation). Therefore, an AI-generated solution does not make further human work on a solved problem entirely meaningless. The task can shift from "solve the problem" to "find an alternative, more illuminating or more efficient solution."

Verifying and understanding an AI solution can itself require substantial mathematical thinking. The case of Grigori Perelman seems especially instructive here. His solution of the Poincaré conjecture was followed by years of verification, reconstruction, and

exposition by other mathematicians, as well as considerable controversy about contribution and credit. The same issues might arise with an AI-generated solution. Indeed, an AI system could be available on demand to explain individual steps repeatedly and in different ways, including by comparison with alternative approaches, although those explanations would of course also have to be checked.

For all these reasons, I do not see AI solutions as the end of mathematical thinking. Additionally, AI models themselves open new directions for mathematical inquiry—AI as an object of research. One highly relevant question is, for example: how is it possible that a system operating on tokenized representations can arrive at correct solutions to mathematical problems? What if mathematical analyses can be modeled through next-token prediction based on patterns extracted from existing data, much as language production is modeled in LLMs (something that, by the way, makes linguists unhappy because it does not align with linguistic units and habits of analysis)?

I also wonder whether the present discussion could be broadened further by including additional perspectives. It would be particularly interesting to hear Grigori Perelman's view. I do not see his rejections of the Fields Medal and the Clay Mathematics Institute's million-dollar prize as a rejection of science. He rarely speaks publicly, particularly through journalists, but who knows: perhaps an invitation from fellow Fields medallists might persuade him to comment on the role of AI in mathematics.

The Advisory Group on Mathematics and Artificial Intelligence (Institute for Advanced Study, Princeton), announced just yesterday, September 21, was established to advise AI companies on their interactions with the mathematical community. Nevertheless, it seems to me that the exchange should be in both directions. If mathematicians are discussing what AI may do to mathematics, perhaps some of the people who developed modern AI should be invited to contribute their perspectives to this discussion. It seems to me that Ilya Sutskever is the obvious person to turn to for an opinion.

For non-mathematicians

More about Grigori Perelman at: <https://www.youtube.com/watch?v=Ng1W2KUHI2s>

The original language of this YouTube documentary is Russian, and it also has English subtitles. If they are not automatically displayed, you have to select them. I advise you to watch and discuss the video with your children. Perelman's story is worth this, believe me.

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